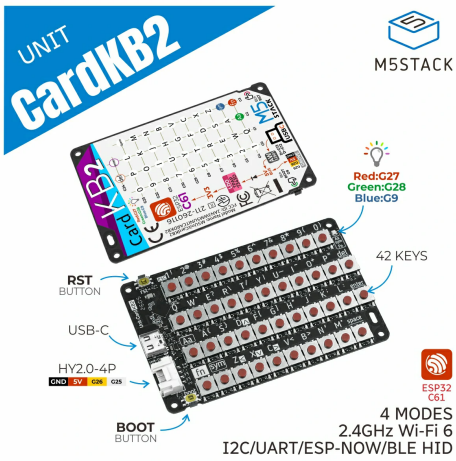
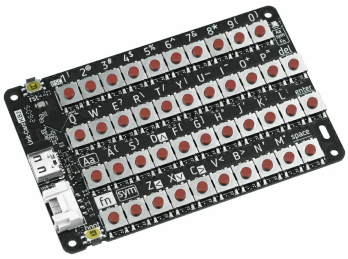
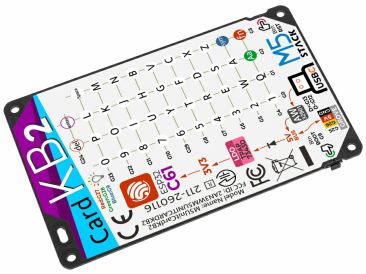
Unit CardKB2

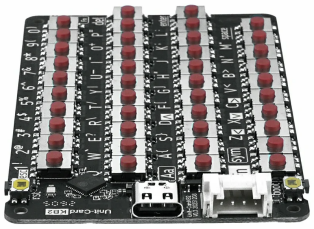
SKU:U215

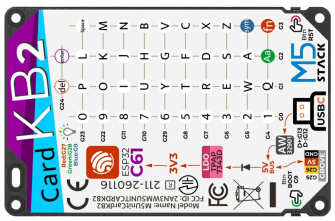


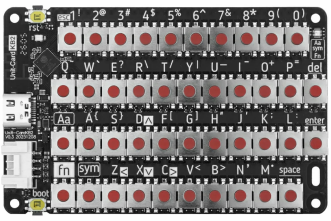










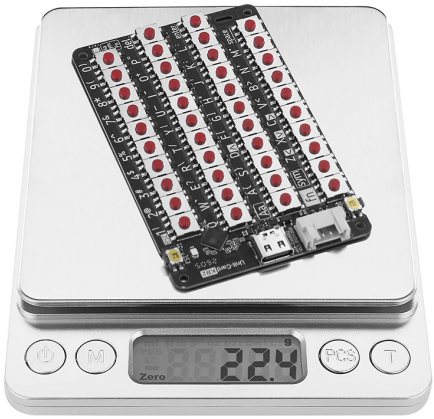












Description

Unit CardKB2 is a card-sized 42-key portable keyboard input unit. Compact and lightweight, it is designed for everyday carry and embedded integration. Powered by the ESP32-C61HF4, it supports 2.4 GHz Wi-Fi 6. The preloaded firmware supports four communication modes — I2C, UART, BLE HID, and ESP-NOW — for flexible connectivity with host devices. An onboard HY2.0-4P connector enables communication in I2C/UART mode, while a USB Type-C port provides power supply and firmware flashing. Additional features include an RGB LED status indicator, a Reset button, a Boot button, and built-in input overvoltage protection, making it well-suited for wireless input, interactive control, and a wide range of portable applications.

Features

- Equipped with ESP32-C61HF4, supporting Wi-Fi 6 / BLE wireless communication
- Full keyboard input with support for multiple key combinations (single press, Sym, Fn)
- Onboard RGB LED (XL-1615RGBC-RF) for keyboard status indication
- Reserved M2 mounting holes for easy application integration
- Power Supply:
  - USB Type-C
  - DC 5V

Includes

- 1 x Unit CardKB2
- 1 x HY2.0-4P Grove Cable (20cm)

Applications

- Text input for portable devices
- Human-machine interaction in embedded systems
- Wireless control terminal for IoT devices
- Educational programming and maker projects

Specifications

Specification	Parameter
SoC	ESP32-C61HF4 @ RISC-V 32-bit single-core, 120 MHz
Flash	4MB
Wi-Fi	2.4 GHz Wi-Fi 6
Keys	42-key keyboard, 1x Reset button, 1x Boot button
USB Type-C	Power supply / Firmware flashing, default firmware log output: 115200bps @ 8N1
RGB LED	1x RGB LED
Expansion Interface	1x HY2.0-4P
Standby Power	5V@19.31mA (powered via Grove interface)
Product Size	84.7 x 54.3 x 1.0mm
Product Weight	22.4g
Package Size	170.0 x 120.0 x 8.0mm
Gross Weight	28.9g

Learn

Key Description

- **Aa Key (Caps Toggle)**:
  - **Single tap**: Enables one-shot caps lock — types one uppercase character, then reverts to lowercase automatically.
  - **Double tap**: Enables caps lock — tap or double-tap again to disengage and return to lowercase.
  - **Hold**: Maintains uppercase output while held; reverts to lowercase upon release.
  - The **green LED lights up** when caps is active and turns off when disabled.
- **Sym Key (Symbol Key)**: Single-click to **activate** character input; click again to deactivate.
- **Fn Key (Function Key)**: Used for mode switching and combined functions. When Fn is pressed, the **red indicator light turns on**.

The following only works in BLE HID mode:

- **Fn + 1**: Outputs Esc
- **Fn + D**: Cursor up (↑)
- **Fn + X**: Cursor down (↓)
- **Fn + Z**: Cursor left (←)
- **Fn + C**: Cursor right (→)
- **Mode Switching (Fn + Sym + Number Key)**: The selected mode is saved automatically and persists after power cycling.
  - **Fn + Sym + 1**: Switch to I2C mode (factory default) — white LED blinks 1 time
  - **Fn + Sym + 2**: Switch to UART mode — white LED blinks 2 times
  - **Fn + Sym + 3**: Switch to ESP-NOW broadcast mode — white LED blinks 3 times
  - **Fn + Sym + 4**: Switch to BLE HID mode — white LED blinks 4 times

BLE HID Mode

Unit CardKB2 can function as a wireless keyboard and connect to any BLE HID-compatible device. Follow the steps below:

1. Press **Fn + Sym + 4** to switch to BLE HID mode.
2. On your phone / tablet / computer, scan for nearby Bluetooth devices. The device will appear as `CardKB2-xxxx` (where `xxxx` represents the last four digits of the MAC address).

3. Pair and connect — the unit is now ready to use as a wireless keyboard.

For usage instructions for other modes (UART / I2C / ESP-NOW), please refer to the Unit CardKB2 User Manual & Registers.

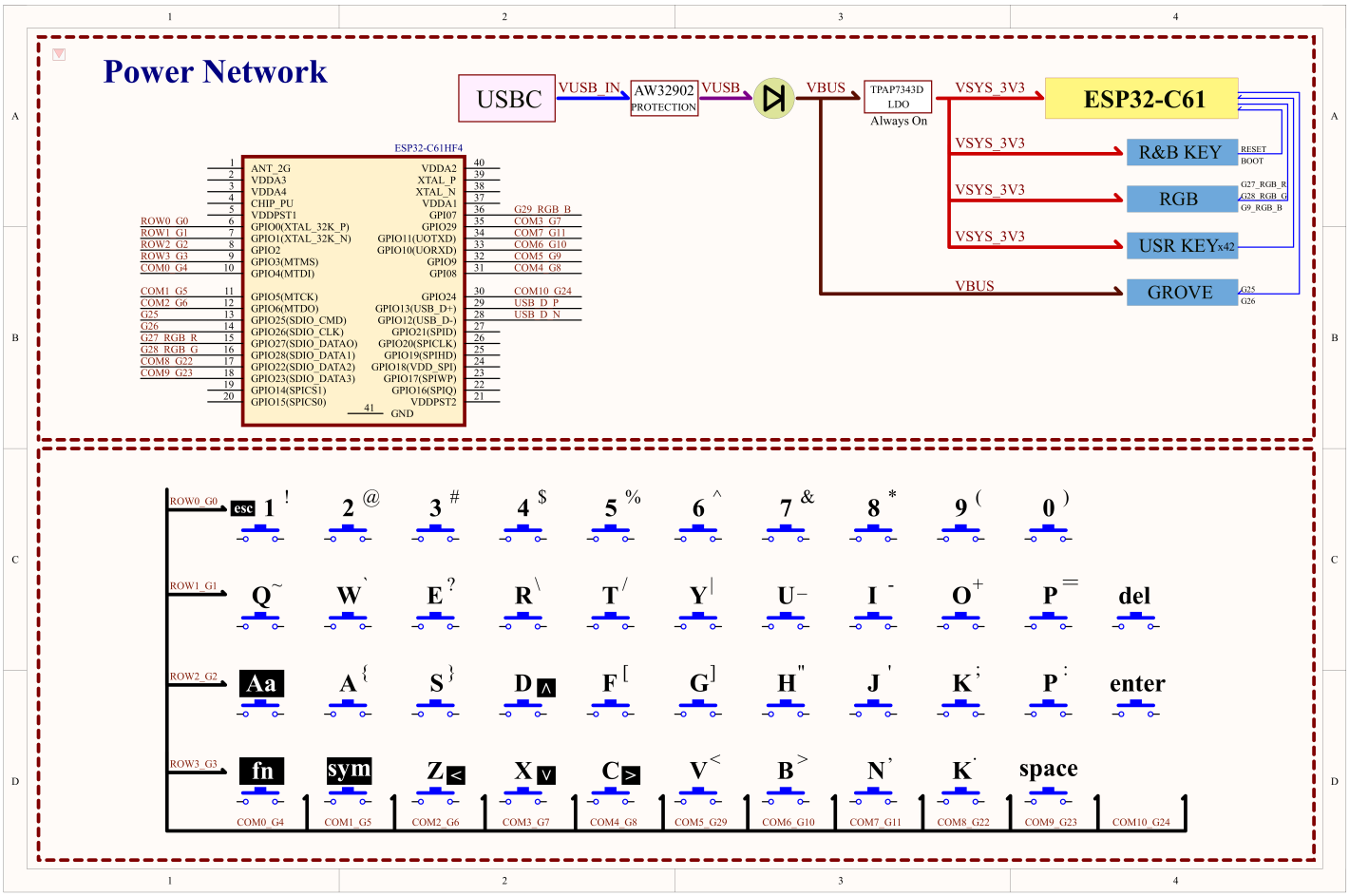
PinMap

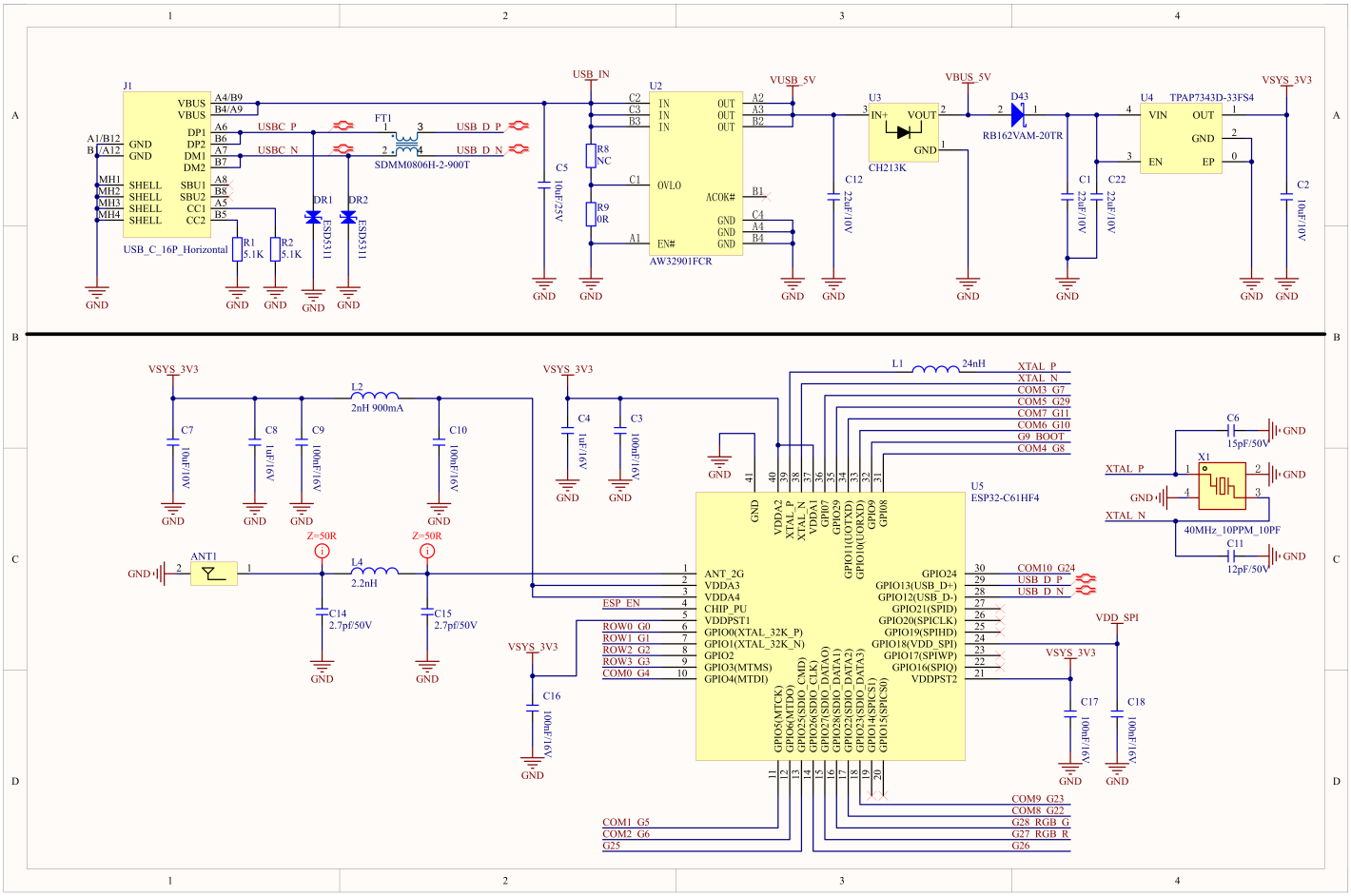
Unit CardKB2

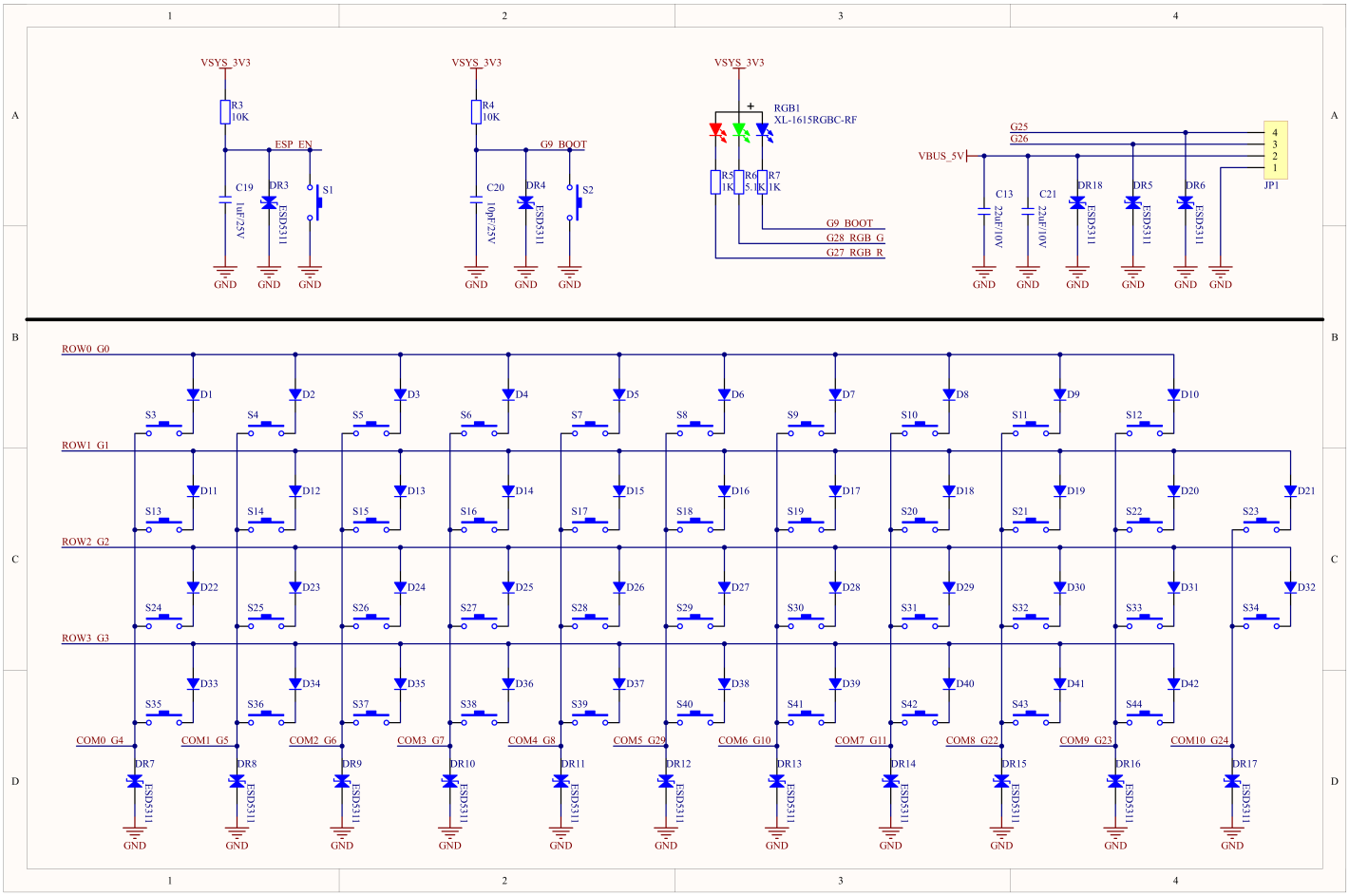
HY2.0-4P	Black	Red	Yellow	White
PORT.CUSTOM	GND	5V	SDA / UART_RX / G26	SCL / UART_TX / G25

Schematics

- Unit CardKB2 Schematics PDF

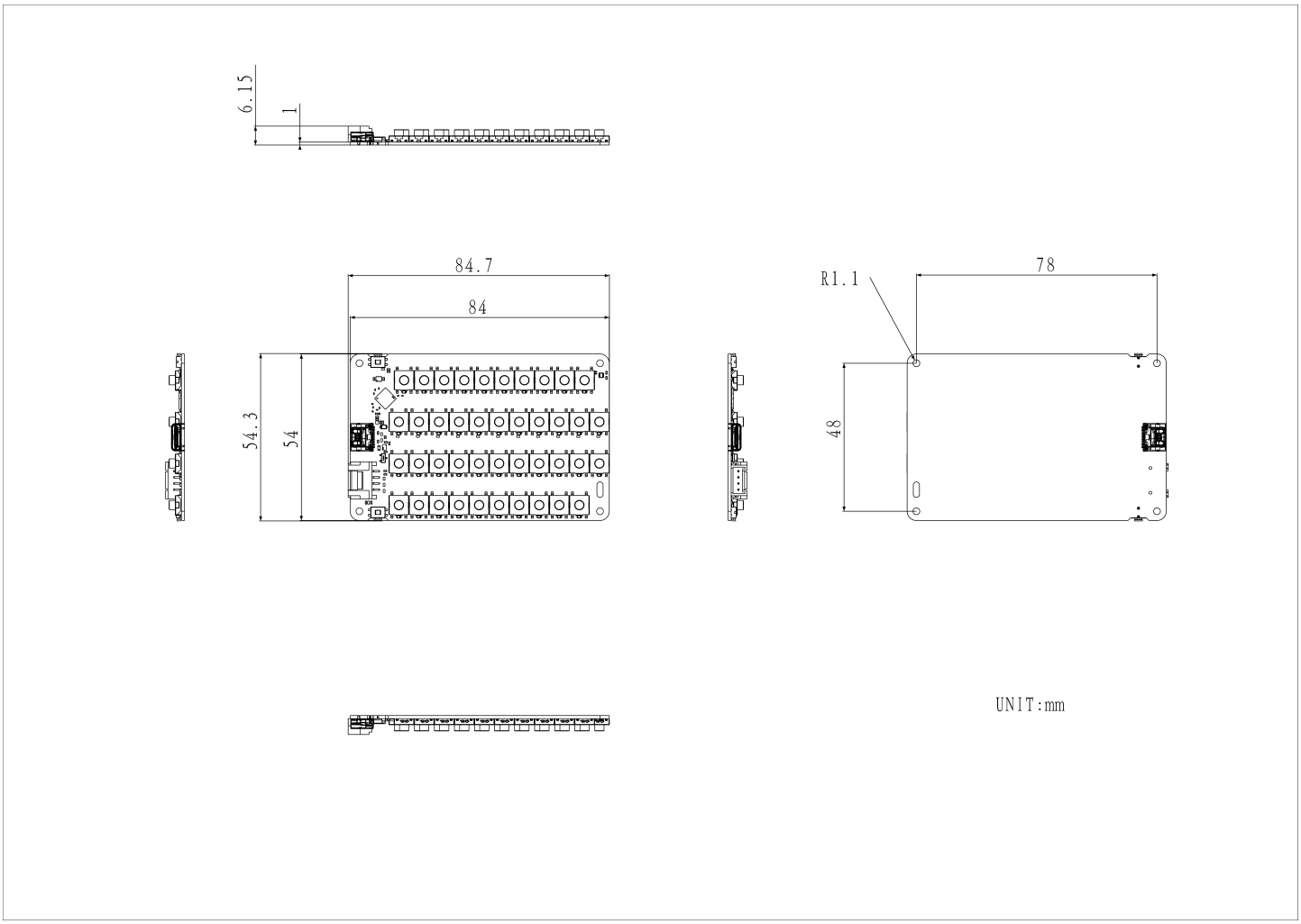



J1
USB_C_16P_Horizontal
USBC_P
USBC_N
FT1
SDMM0806H-2-900T
USB_D_P
USB_D_N
DR1
DR2
ESD5311
R1 5.1K
R2 5.1K
USB_IN
U2
AW32901FCR
R8 NC
R9 0R
C5 10uF/25V
VUSB_5V
U3
CH213K
VBUS_5V
D43
RB162VAM-20TR
U4
TPAP7343D-33FS4
VSYS_3V3
C12 22uF/10V
C1 22uF/10V
C22 22uF/10V
C2 10uF/10V
GND
L2
2nH 900mA
C7 10uF/10V
C8 1uF/16V
C9 100nF/16V
C10 100nF/16V
C4 1uF/16V
C3 100nF/16V
L1 24nH
ANT1
Z=50R
L4 2.2nH
C14 2.7pf/50V
C15 2.7pf/50V
C16 100nF/16V
U5
ESP32-C61HF4
ANT_2G
VDDA3
VDDA4
CHIP_PU
VDDPST1
GPIO0(XTAL_32K_P)
GPIO1(XTAL_32K_N)
GPIO2
GPIO3(MTMS)
GPIO4(MTDI)
GPIO5(MTCK)
GPIO6(MTDO)
GPIO24
GPIO13(USB_D+)
GPIO12(USB_D-)
GPIO21(SPID)
GPIO20(SPICLK)
GPIO19(SPIHD)
GPIO18(VDD_SPI)
GPIO17(SPIWP)
GPIO16(SPIQ)
VDDPST2
XTAL_P
XTAL_N
X1
40MHz_10PPM_10PF
C6 15pF/50V
C11 12pF/50V
VDD_SPI
C17 100nF/16V
C18 100nF/16V
ESP_EN
ROW0_G0
ROW1_G1
ROW2_G2
ROW3_G3
COM0_G4
COM1_G5
COM2_G6
G25
COM3_G7
COM5_G29
COM7_G11
COM6_G10
G9_BOOT
COM4_G8
COM10_G24
USB_D_P
USB_D_N
COM9_G23
COM8_G22
G28_RGB_G
G27_RGB_R
G26


R3 10K
ESP_EN
C19 1uF/25V
DR3
S1
R4 10K
G9_BOOT
C20 10pF/25V
DR4
S2
RGB1
XL-1615RGBC-RF
R5 1K
R6 5.1K
R7 1K
G9_BOOT
G28_RGB_G
G27_RGB_R
G25
G26
JP1
C13 22uF/10V
C21 22uF/10V
DR18
DR5
DR6
ROW0_G0
ROW1_G1
ROW2_G2
ROW3_G3
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S43
S44
COM0_G4
COM1_G5
COM2_G6
COM3_G7
COM4_G8
COM5_G29
COM6_G10
COM7_G11
COM8_G22
COM9_G23
COM10_G24
DR7
DR8
DR9
DR10
DR11
DR12
DR13
DR14
DR15
DR16
DR17

Model Size

Unit CardKB2 Model Size PDF



Datasheets

ESP32-C61HF4

Softwares

Arduino

- Unit CardKB2 Arduino Tutorial
- Unit CardKB2 Arduino Library

UiFlow2

coming soon...

Internal Firmware

- Unit CardKB2 Internal Firmware

Protocol

- Unit CardKB2 User Manual & Registers PDF

Video

- Unit CardKB2 Product Introduction and Feature Demonstration

U215-UnitCardKB2_video_EN.mp4